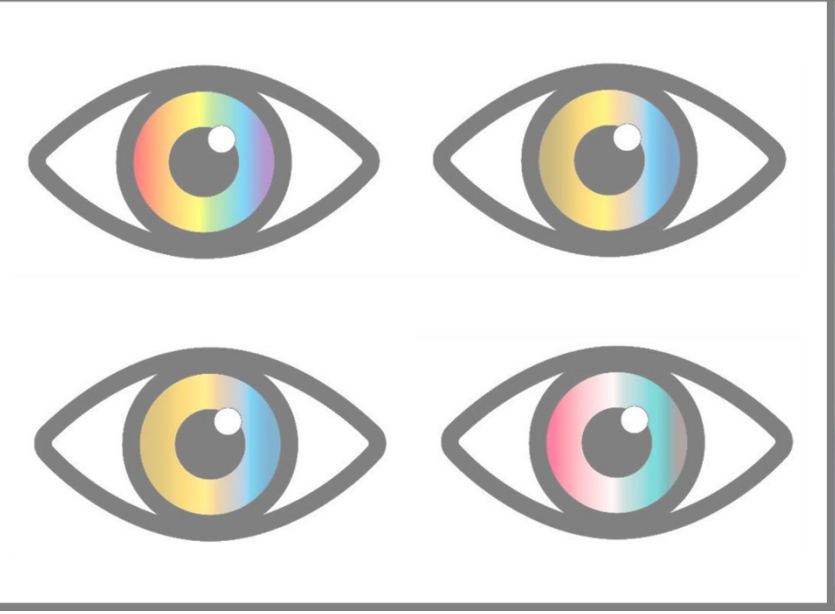

Colour Vision Deficiency in Chemistry: Challenges and Practical Solutions

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Dalhousie University, Department of Chemistry

Canadian Chemistry Conference and Exhibition
June 8th, 2023

The Colours Project

The Team

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Funding

- LearnSci/CLEAR21 Grant
- Dalhousie University
  - Department of Chemistry
  - Centre for Learning and Teaching
  - Office of Accessibility and Accommodation

Colour Vision Deficiency (CVD)

- CVD affects ~9% of the world's population
- Difficulty or inability to distinguish certain colours
- Colour vision is dependent on **cones**

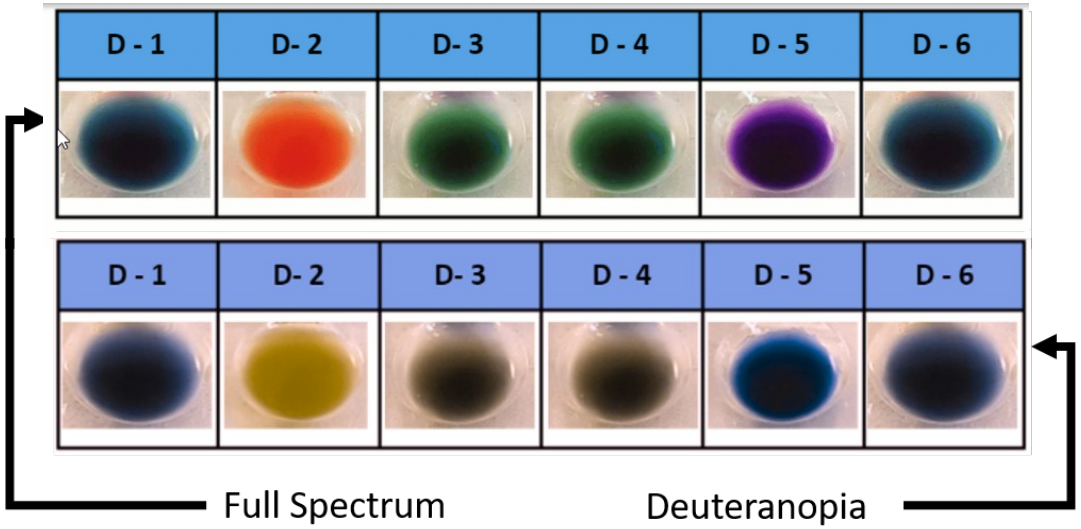
CVD Type	Lack of Pigment or Shift in Receptor	Commonly Confused Colours
Protanopia	Red (Long λ)	Red, Green, Brown, Purple
Deuteranopia	Green (Medium λ)	Red, Green, Blue, Purple
Tritanopia	Blue (Short λ)	Blue, Green, Yellow

(1) Colour Blind Awareness, (2) McIntyre, (3) Chromatic Vision Simulator

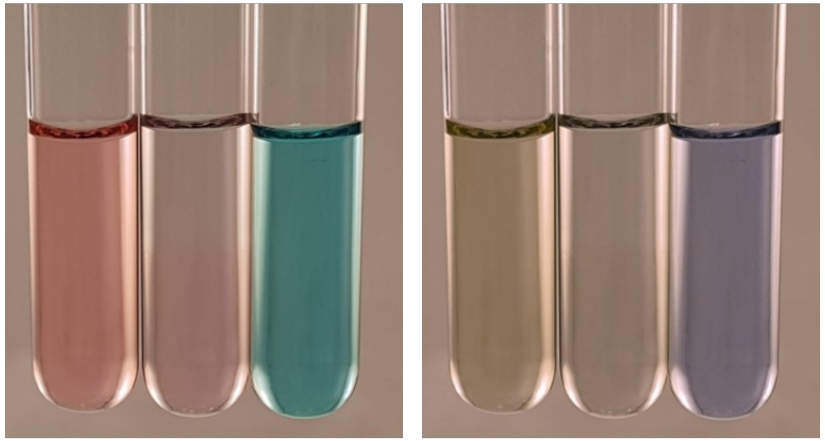
CVD in the First-Year Chemistry Laboratory (1)

- Potentially ~105 students in our class and ~3 teaching staff
- 10 out of 16 First-Year Chemistry experiments have some aspect of colour-based data/observations

Universal Indicator



Column Chromatography

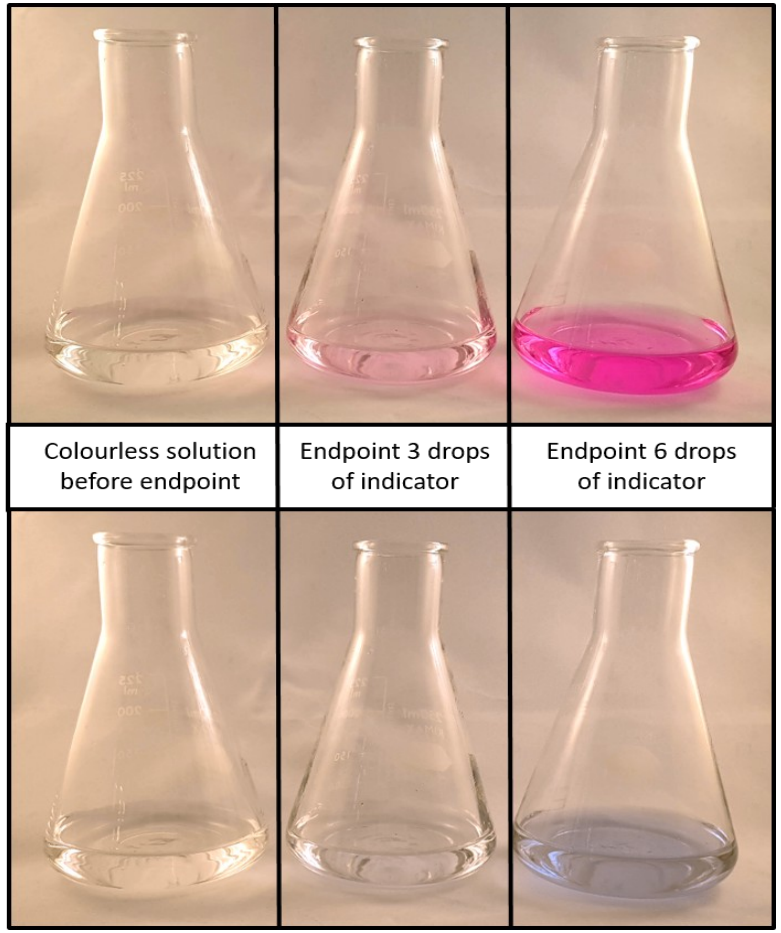


Full Spectrum Deuteranopia

(5) Coblis Colour Blindness Simulator

CVD in the First-Year Chemistry Laboratory (2)

Full Spectrum



Deuteranopia

"**I feel like I am cheating** if I ask you to confirm that I have assigned my colours correctly."

- On the left are two portrayals of titration endpoints
- Dramatic difference in noticing the endpoint
- How can we help?

(5) Coblis Colour Blindness Simulator

Strategies for Building CVD Friendly Content (1)

- Don't eliminate colour entirely, instead…
- Consider your colour palette carefully
- Simulate what this colour palette may look like for others to ensure the colours you selected appear unique
- Avoid using colour as the sole communicator of information (pair with text, pattern, or animation)

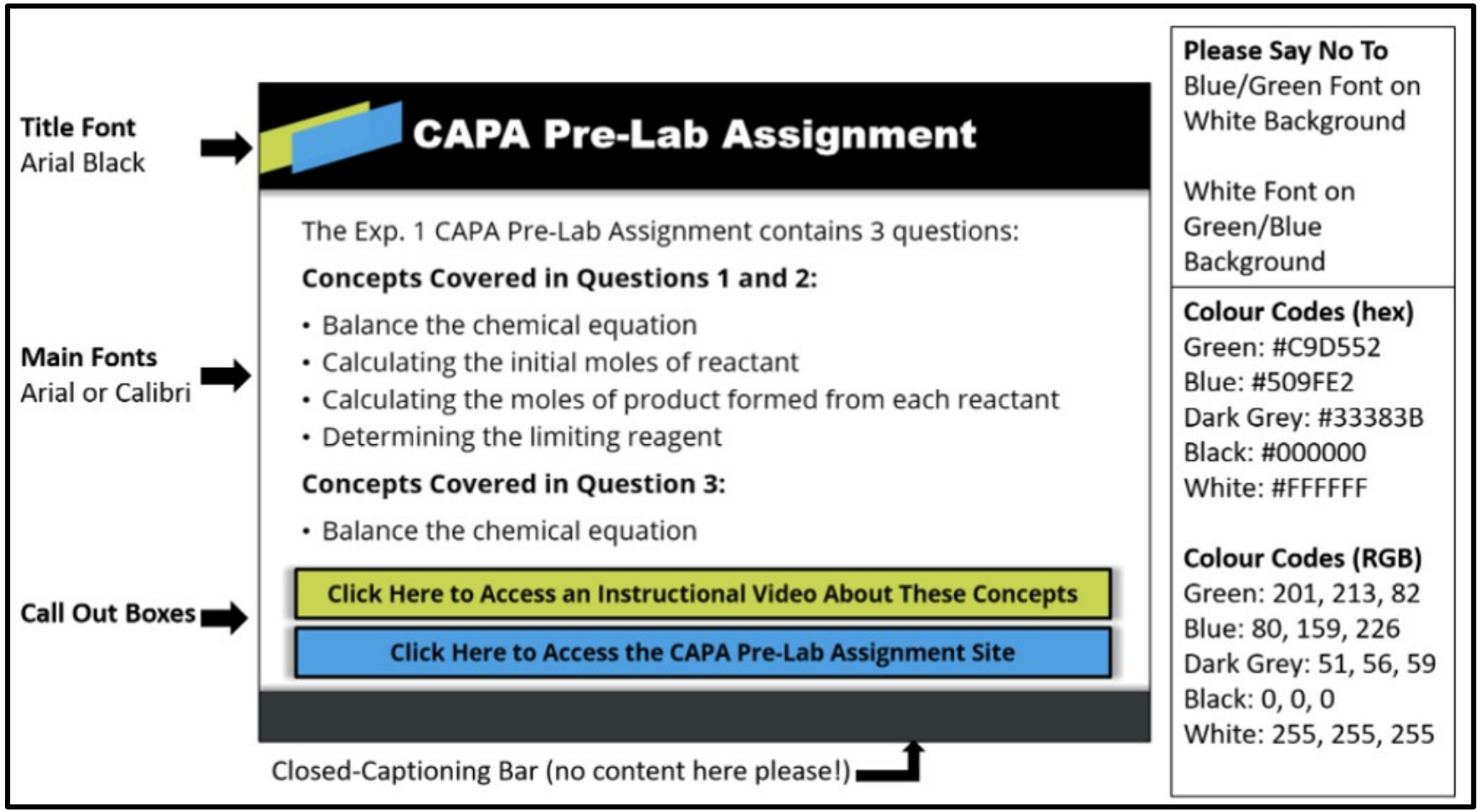
Colour Palette

(4) Coloring for Colorblindness

Strategies for Building CVD Friendly Content (2)

Make records of:

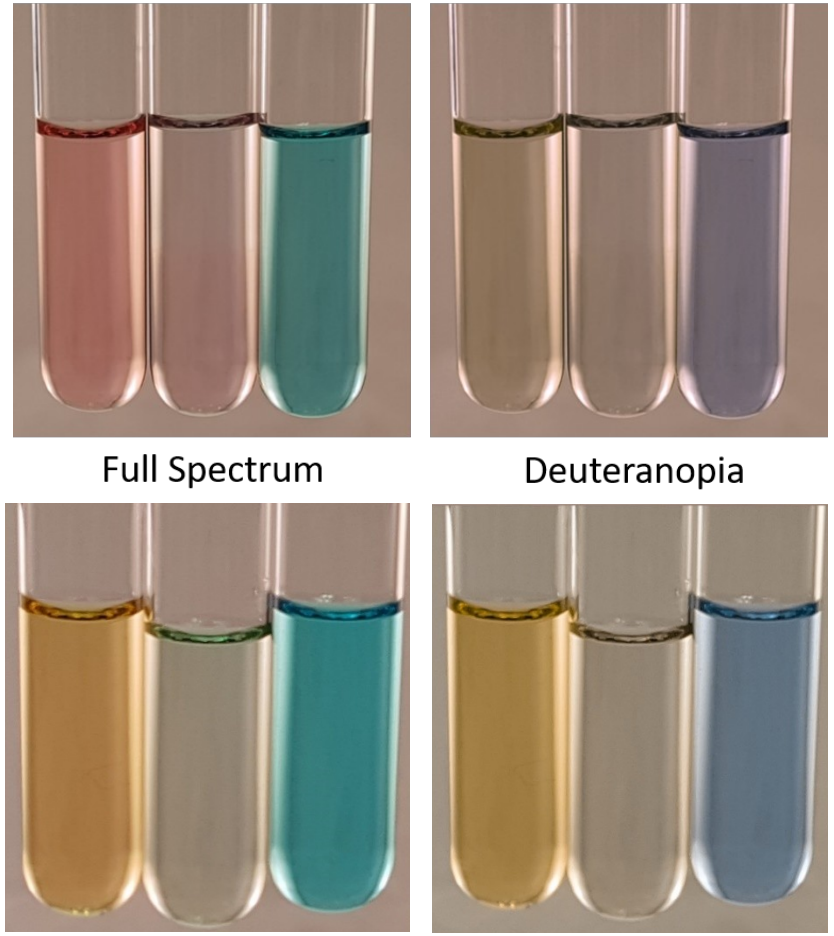
- Colour choices and Codes
- Colour combinations to avoid which generate poor contrast
- Share with your team if you are co-developing content



(4) Coloring for Colorblindness, (5) Coblis Colour Blindness Simulator, (6) WebAIM

Supporting Students and Educators (1)

Column Chromatography



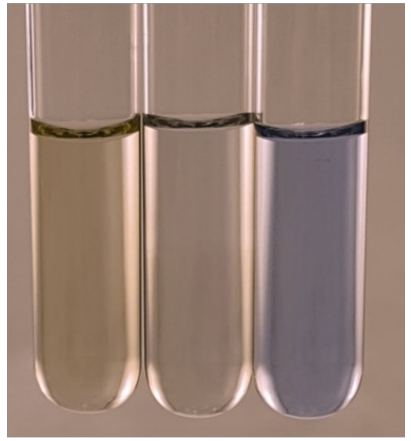
- **Student Facing:** Same experiment, option for CVD Friendly unknown

- **Lab Manual Text:** If you or a teammate suffers from colour vision deficiency (CVD), feel free to ask the teaching team for an unknown with a dye colour mixture that supports red/green CVD or blue/yellow CVD.

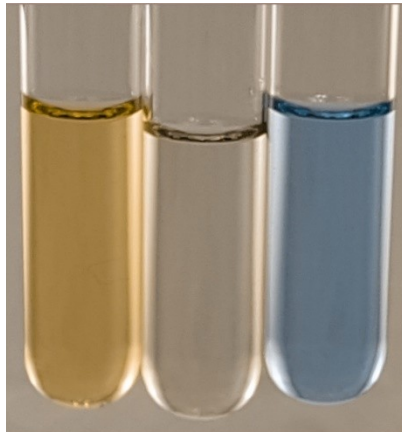
Supporting Students and Educators (2)

Column Chromatography

Full Spectrum



Deuteranopia



- **Teaching Team Facing:** Custom pocket-sized note card linking unknowns to colours present (red, green, blue, yellow)

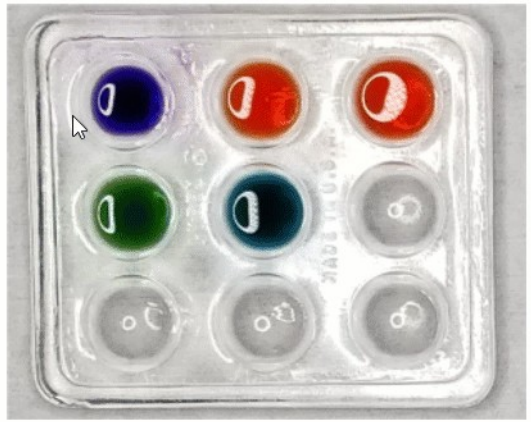
		Unknown Numbers		
Dye 1	Dye 2	Main	CVD Support	CVD Type
Red	Blue	2-digit unknown number	3-digit unknown number	Blue/Yellow CVD (Tritanopic)
Red	Green	2-digit unknown number		
Yellow	Blue	2-digit unknown number	3-digit unknown number	Red/Green CVD (Protanopic/ Deuteranopic)
Yellow	Green	2-digit unknown number		

Supporting Students and Educators (3)

Teaching Team Training Modules

Helpful Tips – Part B (3)

- **Phenolphthalein** typically causes no problems
- **Metacresol purple** may present a brown colour for neutral solutions (this is OK)
- **Universal indicator** may require additional indicator to resolve weak base (blue) and strong base (purple) unknowns

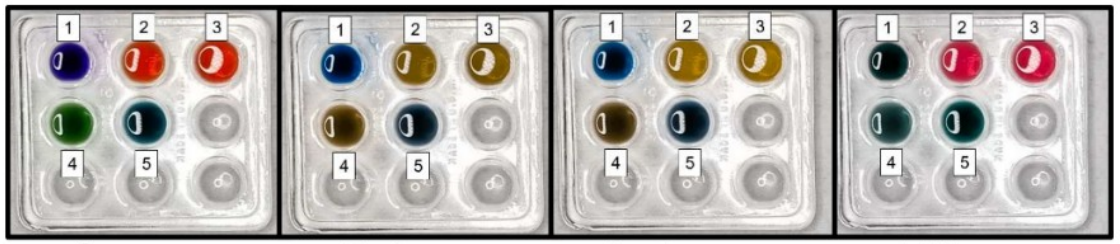


Universal Indicator
Hover over the pH paper to confirm colour identification

Helpful Tips – Part B (4)

- Lab is reliant on full colour vision

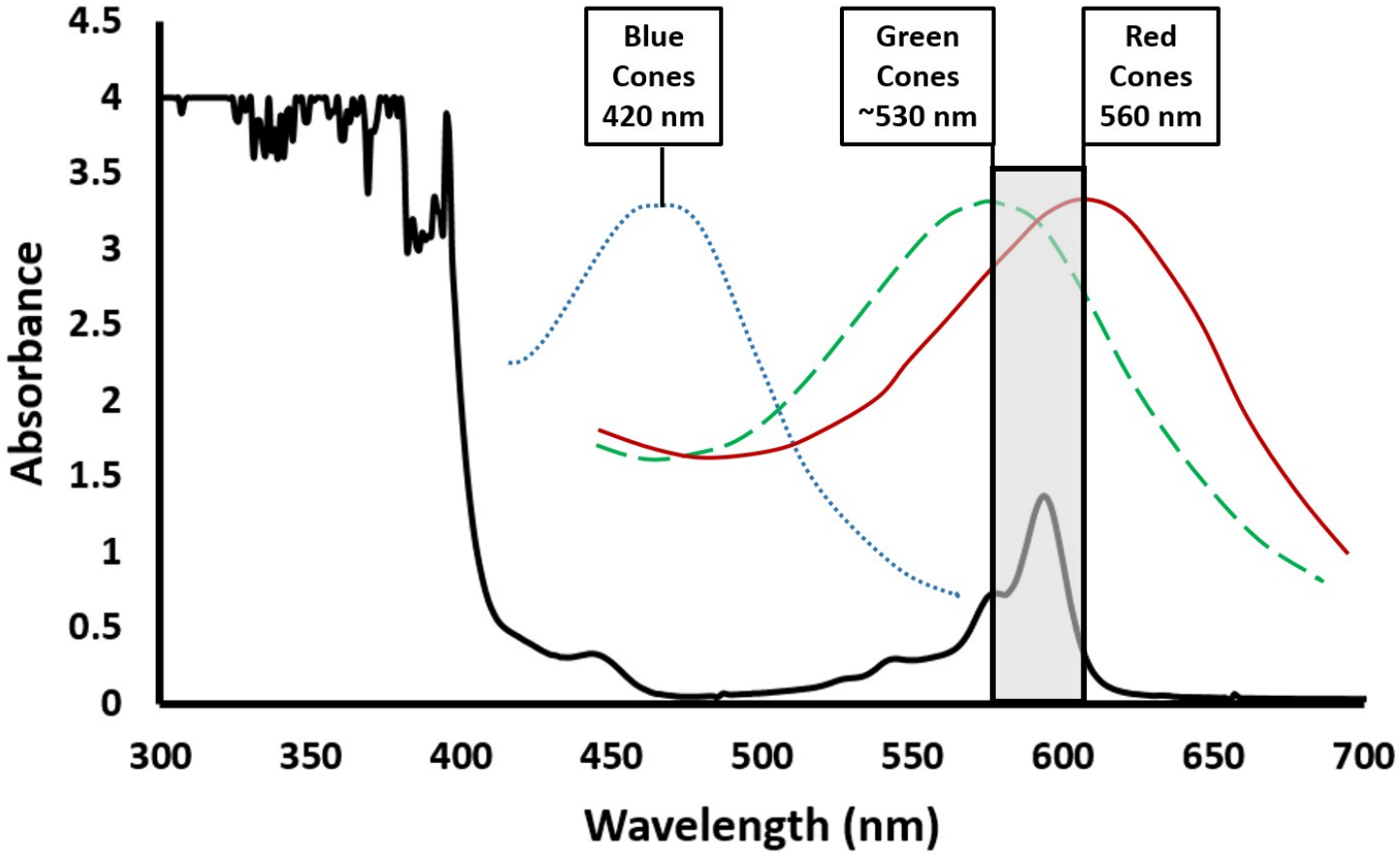
- Colour Vision Deficient (CVD) students are welcome to ask for assistance



Common | Protanope | Deuteranope | Tritanope

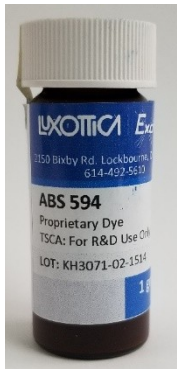
Commercially Available CVD Filter – EnChroma

- Deuteranopia: missing green cones
- EnChroma Glasses “block” wavelengths



(2) McIntyre, (5) EnChroma, (6) We Are Colour Blind

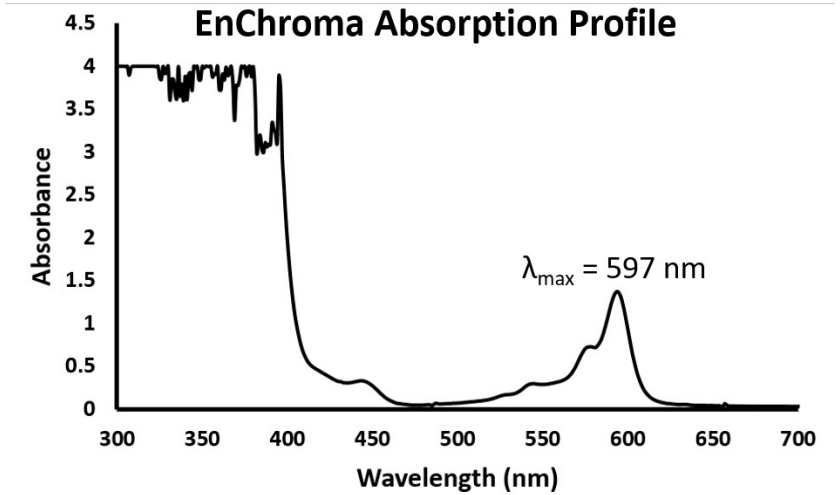
Preparing in-house CVD filter

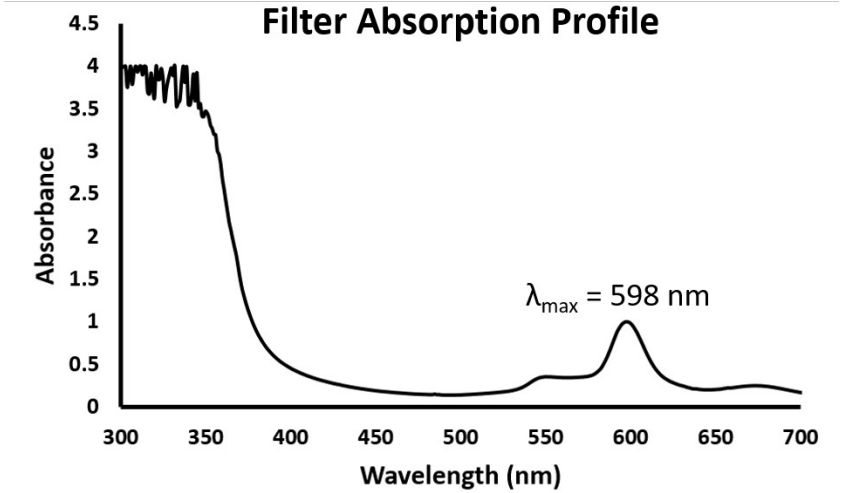


- **Dye:**
  - ~10 mg ABS 594 (Luxottica)

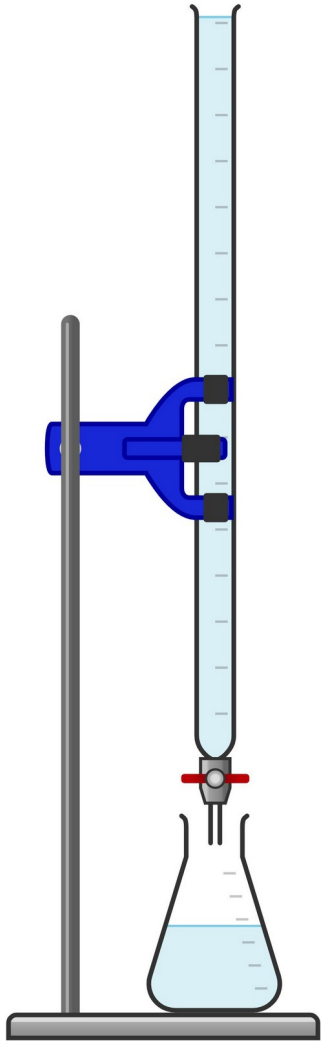
- **Solvent:**
  - 4 mL acrylic polyurethane

- **Casting:**
  - Drop Coat solution on Clean/Dry Glass Lens
  - Cure for 24 Hrs





How do we Quantitate Personal Colour Experience?

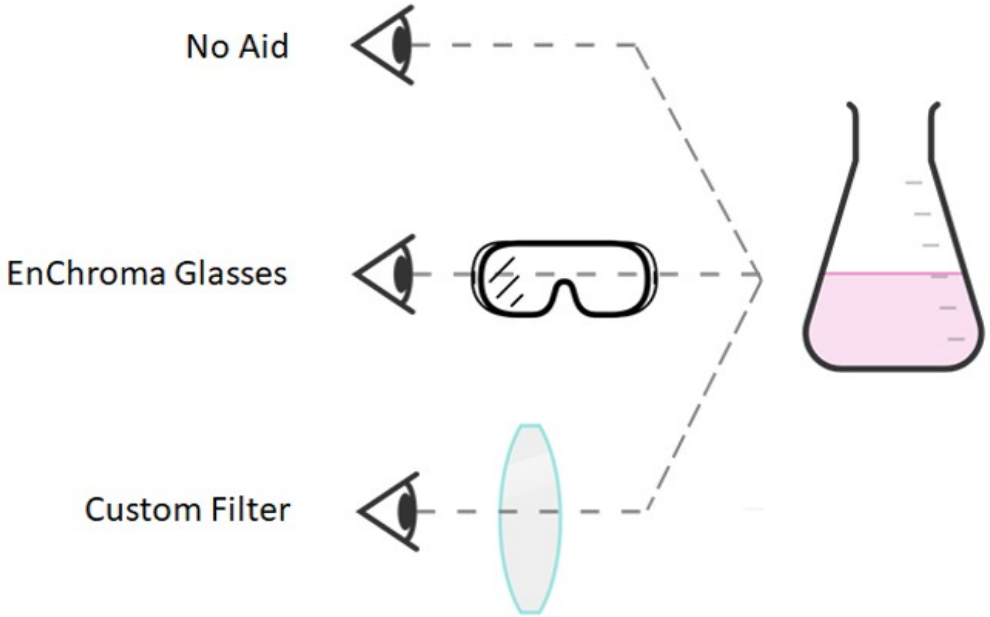


Potassium hydrogen phthalate (KHP) was titrated against standardized Sodium Hydroxide (NaOH)

~0.1 M NaOH
(standardized with pH probe)

~0.5g KHP
3 drops phenolphthalein (0.1%)
25 mL distilled water

- Two groups of individuals:
  - CVD: n = 2
  - Control: n = 2
- Order of titrating was randomized
- Mass of KHP unknown during experiment



Results

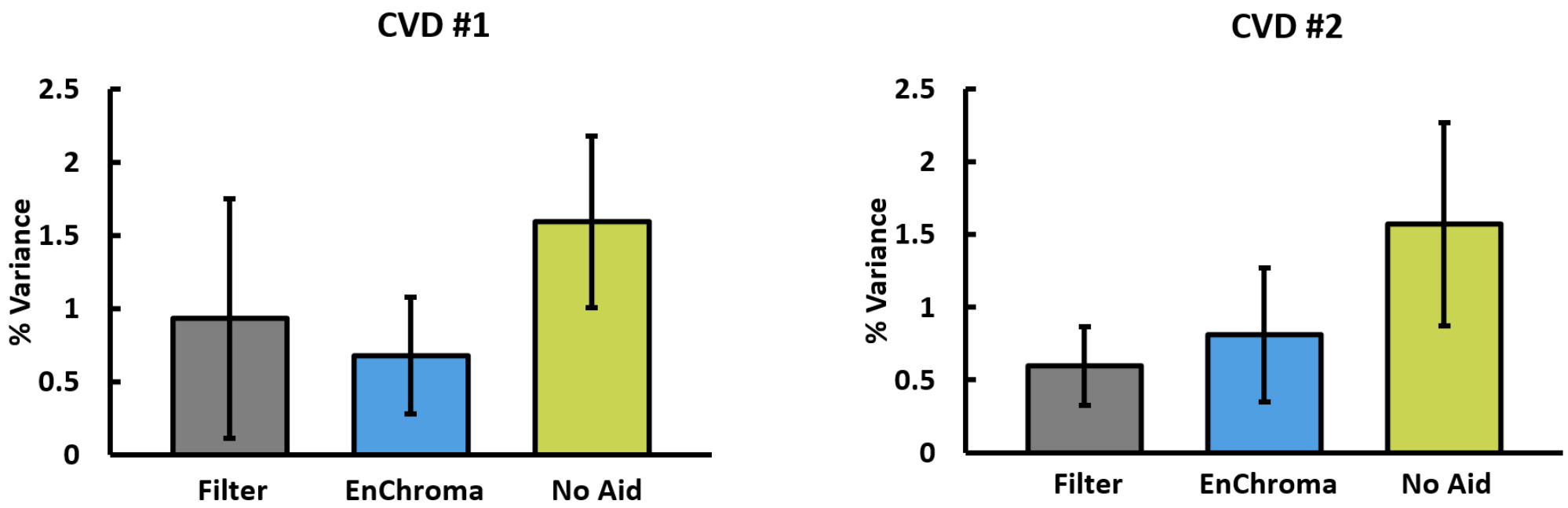


Figure 3: Comparison of % Variance in Moles of KHP From Actual Determined With Filter, EnChroma Glasses, and No Aid for CVD Individuals

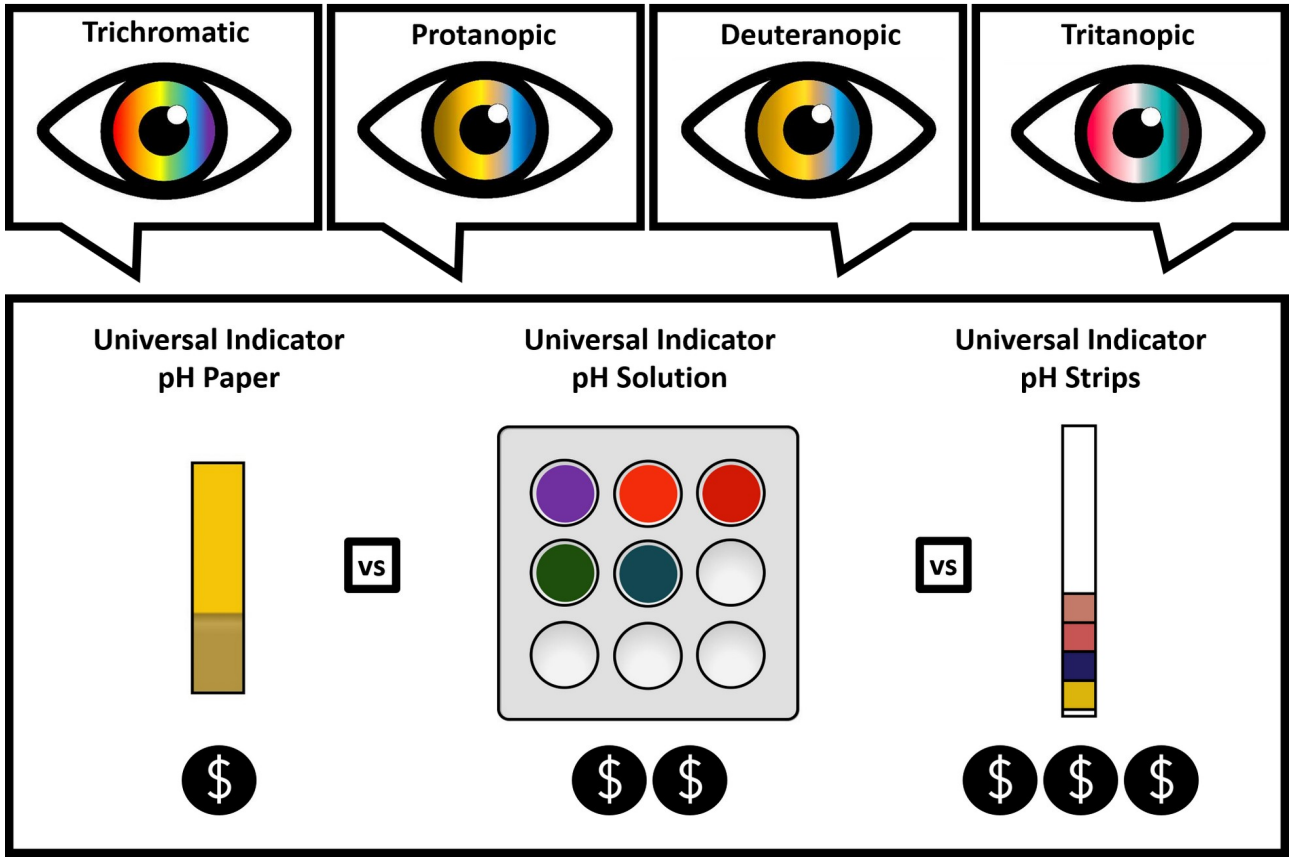
- CVD individuals observed similar visual benefits using the filter as compared to EnChroma glasses
- CVD individuals reported seeing the short-lived splotches of pink solution that signal the endpoint is near with the filter and EnChroma glasses

Students as Partners: Other Works

How Effective Are Indicators for Individuals with Colour Vision Deficiency?

N. J. Roberts, T. Hynes, D. Stacey, J. L. MacDonald (*ChemRxiv*, **2023**)

DOI: [10.26434/chemrxiv-2023-6g77r](https://doi.org/10.26434/chemrxiv-2023-6g77r)



Students as Partners: Other Works

Coming in 2024…an Open Educational Resource!

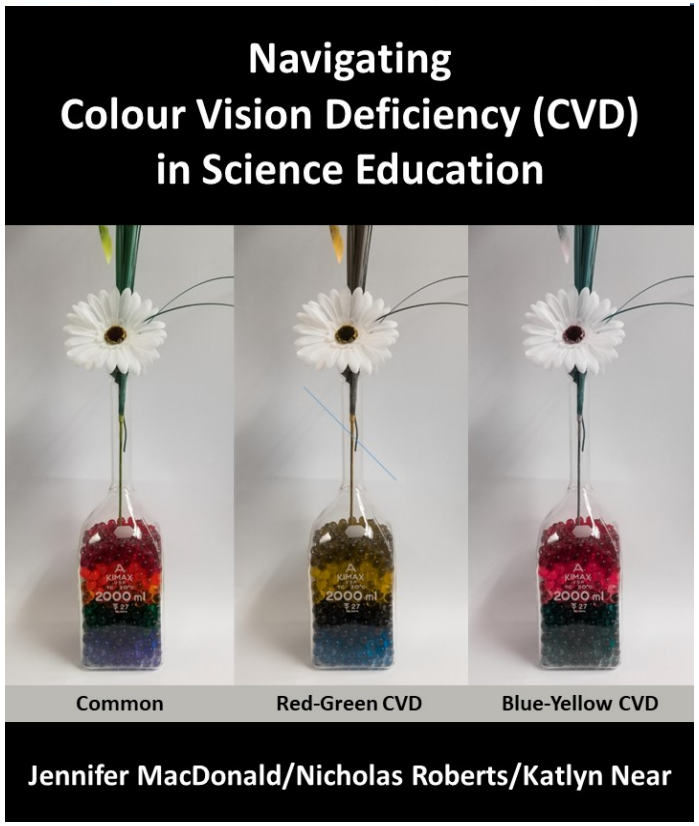
PB PRESSBOOKS

Navigating Colour Vision Deficiency (CVD) in Science Education

J. L. MacDonald, N. J. Roberts, K. G. Near

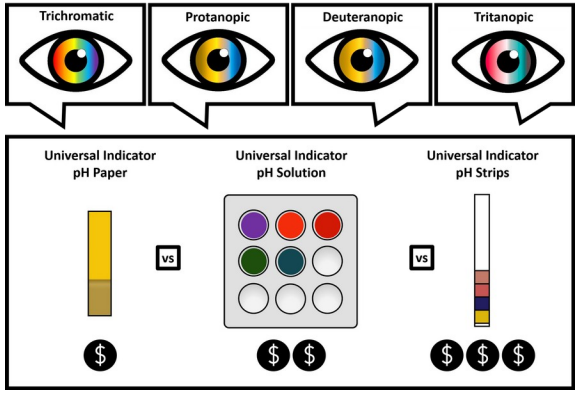
Add yourself to the mailing list to be notified when the OER is released.

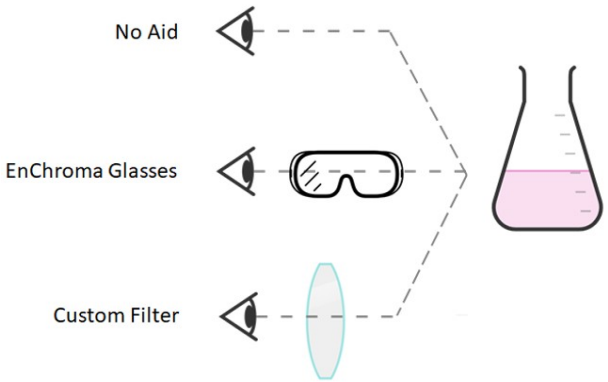
tinyurl.com/ColoursOER

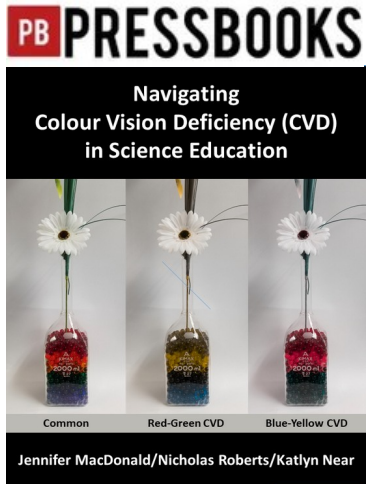


Creating Community

Colour Palette







Awareness, Access, Inclusion and Belonging

Exclusionary Language	Inclusionary Language
"I feel like I am cheating..."	"Our team would like to opt for the colour-blind friendly unknown please."

References & Contact

References

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7. EnChroma Eyewear for Color Blindness https://enchroma.com/ (accessed Feb 11, 2023).

Feel free to contact us!

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